

4.1 Answer each of the following:

- a. Suppose that a simple regression has quantities $N = 20$, $\sum y_i^2 = 7825.94$, $\bar{y} = 19.21$, and $SSR = 375.47$, find R^2 .
- b. Suppose that a simple regression has quantities $R^2 = 0.7911$, $SST = 725.94$, and $N = 20$, find $\hat{\sigma}^2$.
- c. Suppose that a simple regression has quantities $\sum (y_i - \bar{y})^2 = 631.63$ and $\sum \hat{e}_i^2 = 182.85$, find R^2 .

$$\begin{aligned} a. \quad SST &= \sum y_i^2 - N \bar{y}^2 = 7825.94 - 20(19.21)^2 \\ &= 7825.94 - 20(369.0241) \\ &= 445.458 \end{aligned}$$

$$R^2 = \frac{SSR}{SST} = \frac{375.47}{445.458} \approx 0.8429$$

$$b. \quad R^2 = SST(1 - R^2)$$

$$\Rightarrow SSE = SST(1 - R^2) = 725.94(1 - 0.7911) \approx 151.6489$$

$$\hat{\sigma}^2 = \frac{SSE}{N-2} = \frac{151.6489}{20-2} \approx 8.4249$$

$$c. \quad SST = \sum (y_i - \bar{y})^2 = 631.63$$

$$SSE = \sum \hat{e}_i^2 = 182.85$$

$$R^2 = 1 - \frac{SSE}{SST}$$

$$= 1 - \frac{182.85}{631.63}$$

$$= 1 - 0.2895 \approx 0.7105$$

4.3 We have five observations on x and y . They are $x_i = 3, 2, 1, -1, 0$ with corresponding y values $y_i = 4, 2, 3, 1, 0$. The fitted least squares line is $\hat{y}_i = 1.2 + 0.8x_i$, the sum of squared least squares residuals is $\sum_{i=1}^5 \hat{e}_i^2 = 3.6$, $\sum_{i=1}^5 (x_i - \bar{x})^2 = 10$, and $\sum_{i=1}^5 (y_i - \bar{y})^2 = 10$. Carry out this exercise with a hand calculator. Compute

- a. the predicted value of y for $x_0 = 4$.
- b. the $se(f)$ corresponding to part (a).
- c. a 95% prediction interval for y given $x_0 = 4$.
- d. a 99% prediction interval for y given $x_0 = 4$.
- e. a 95% prediction interval for y given $x = \bar{x}$. Compare the width of this interval to the one computed in part (c).

$$a. \quad \hat{y}_0 = 1.2 + 0.8(4) = 4.4$$

$$\begin{aligned} b. \quad \text{Var}(f) &= \hat{\sigma}^2 \left(1 + \frac{1}{N} + \frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2} \right) \\ &= 1.2 \left(1 + \frac{1}{5} + \frac{(4-1)^2}{10} \right) \\ &= 1.2(1 + 0.2 + 0.9) \\ &= 2.52 \end{aligned}$$

$$se(f) = \sqrt{2.52} \approx 1.5875$$

$$c. \quad t_{3, 0.95} = 3.182$$

$$\begin{aligned} \hat{y}_0 \pm t_c se(f) &= 4.4 \pm 3.182(1.5875) \\ &= 4.4 \pm 5.0514 \end{aligned}$$

$$\therefore 95\% \text{ P.I.} \Rightarrow [-0.6514, 9.4514]$$

$$d. \quad t_{0.99, 3} = 5.841$$

$$\begin{aligned} \hat{y}_0 \pm t_c se(f) &= 4.4 \pm 5.841(1.5875) \\ &= 4.4 \pm 9.2726 \end{aligned}$$

$$\therefore 99\% \text{ P.I.} = [-4.8726, 13.6726]$$

$$e. \quad \hat{y}_0 = 1.2 + 0.8(1) = 2$$

$$\text{Var}(f) = 1.2 \left(1 + \frac{1}{5} + \frac{(1-1)^2}{10} \right)$$

$$= 1.2(1 + 0.2) = 1.44$$

$$se(f) = \sqrt{1.44} = 1.2$$

$$\begin{aligned} \hat{y}_0 \pm t_c se(f) &= 2 \pm 3.182(1.2) \\ &= 2 \pm 3.8184 \end{aligned}$$

$$95\% \text{ P.I.} \Rightarrow [-1.8184, 5.8184]$$

Comparison

$$(c) \quad 9.4514 - (-0.6514) = 10.1028$$

$$(e) \quad 5.8184 - (-1.8184) = 7.6368$$

the width of interval in part (e) is narrower than that in part (c)

when x_0 is closer to $\bar{x}$, $\frac{(x_0 - \bar{x})^2}{\sum (x_i - \bar{x})^2}$ becomes smaller, making standard error becomes smaller.

$\therefore$ The prediction interval in part (e) is narrower.